Rashmi Rama Sushil

Mumbai, India | rashmi.ramasushil@gmail.com | linkedin.com/in/rrsk | rashmirrsdev.github.io/rashmi_portfolio

SUMMARY

Quantitative finance professional with 3 years building and operating risk analytics frameworks at Morgan Stanley Trading Risk Management, now leading the development of an AI-assisted commodity risk dashboard using GitHub Copilot and agent-based AI tooling. Daily exposure to a large, diverse commodity derivatives book spanning crude oil, EU power and natural gas, precious and base metals (LME/CME), agricultural markets, and Asian-listed futures (DCE/SGX). MS Research from IIT Madras (CGPA 9.56/10) with 4 peer-reviewed publications. CFA Level 1 certified. Combines derivatives domain expertise with hands-on AI-assisted software engineering to modernise how risk is analysed, visualised, and communicated to stakeholders.

SKILLS

Technical: Python (pandas, NumPy, SciPy, statsmodels, matplotlib), SQL, Power BI, Excel, Bloomberg Terminal, GitHub Copilot & Agent-Based AI Development (VS Code), Claude, Cursor

Finance: Derivatives Analytics, Greeks (Delta/Gamma/Vega/Theta), P&L Attribution, Risk Sensitivities, Factor Analytics (Beta, HHI), VaR, Stress Testing, Deal Structure Evaluation

Domain: Commodity Derivatives — crude oil, EU power and natural gas, precious & base metals (LME/CME), agricultural, Asian futures (DCE/SGX); Multi-asset Futures & Options; Bloomberg

EXPERIENCE

Morgan Stanley *Jan 2025 – Present*
Senior Associate — Trading Risk Management Mumbai

- Leading the design and development of an AI-assisted commodity risk dashboard — using GitHub Copilot and agent-based AI tooling in VS Code to modernise how the desk visualises forward curves, time spreads, cross-curve basis spreads, and baskets of trades.
- Designing a framework for evaluating potential deal structures, systematically comparing structuring alternatives against their risk and P&L profile.
- Built and operated P&L production and reporting tools; led daily and weekly risk & P&L review forums with Director/MD-level leadership across global commodity desks.
- Performed return attribution and derivatives exposure analytics (Greeks, stress analysis, Sharpe ratio) across multi-asset books with multi-million dollar notional.
- Monitored 20+ instruments daily on Bloomberg, tracking price shocks, vol shifts, and macro catalysts across crude oil, EU power and natural gas, precious and base metals (LME/CME), agricultural markets, and Asian-listed futures (DCE/SGX).
- Developed executive dashboards and risk narratives for Director/MD-level stakeholders, translating quantitative analysis into clear, decision-relevant communication.
- Partnered with traders, quants, and data teams to scope and deliver analytical tools aligned with desk requirements.

Morgan Stanley *Jul 2023 – Dec 2024*
Associate — Trading Risk Management Mumbai

- Built, maintained, and extended 30+ structured risk reporting tools monitoring \$50M–\$300M notional across 5+ global desks; end-to-end platform ownership from data ingestion to stakeholder distribution.
- Developed and validated quantitative models tracking derivatives risk sensitivities (Delta, Vega, Gamma, Theta) and identified portfolio concentration risks through factor analytics (Beta, hedge ratio, HHI).
- Automated reporting and reconciliation workflows in Python, SQL, Power BI, and Excel, reducing manual workload by 40% and improving accuracy across daily risk deliverables.
- Mapped data lineage across multiple upstream systems; built scalable, reusable data pipelines integrating cross-desk position and market data.

IIT Madras *Sep 2020 – Jun 2023*
Teaching Assistant Chennai

- Independently managed and guided 80+ students across graduate-level engineering courses while completing MS Research.

ISRO (Indian Space Research Organisation) *Jun 2017 – Jul 2017*
Research Intern Bangalore

- Analysed 20+ technical papers on inflatable antenna feasibility; designed, built, and tested a functional prototype.

OPEN-SOURCE PROJECTS

Options Risk Analytics Engine github.com/rashmirrsdev/options-risk-engine
Python · Black-Scholes · Greeks · VaR · Stress Testing
BSM pricing engine computing full Greeks (Delta, Gamma, Vega, Theta, Rho) per position. Historical and parametric (delta-normal) VaR at 95%/99%. 10 configurable stress scenarios with P&L impact. Delta concentration flags for dominant-risk identification. 13 unit tests.

Factor Risk Attribution Engine github.com/rashmirrsdev/factor-risk-attribution
Python · Rolling OLS · Beta · Nifty 50 · Return Decomposition
Decomposes 3 years of Nifty 50 portfolio returns into systematic (Beta) and idiosyncratic components using rolling OLS regression. Computes HHI concentration index, sector exposures, and idiosyncratic Sharpe ratio to evaluate stock-selection quality. 5 attribution charts. 12 unit tests.

Options Hedging Strategies Analyser github.com/rashmirrsdev/options-hedging-strategies
Python · Payoff Analysis · Breakeven · INR P&L
Models 6 strategies (unhedged, protective put, covered call, collar, bull/bear spread, long straddle) for a long equity position. Computes exact INR payoffs, max gain/loss, and breakeven points via linear interpolation. Visual risk/cost comparison across all strategies. 17 unit tests.

Metals Hedging Stress Test github.com/rashmirrsdev/metals-hedging-stress
Python · Time Spreads · DV01 · Stress Testing
Stress-tests a metals derivatives hedge across two distinct exposures: front-end slippage risk and back-end holding risk. Liquidity screening via open interest, time spread decomposition, rolling N-day stress statistics (max/p99/p95/min), and DV01/lease-rate sensitivity specific to precious metals financing.

EDUCATION

Indian Institute of Technology Madras (IIT Madras) *Sep 2020 – Jun 2023*
MS Research, Engineering Design Chennai
CGPA: **9.56 / 10** | Research: ML methods for large dataset visualisation and exploration
Publications: Springer 2025 • WCSMO USA 2023 • ASME USA 2022 • ACSMO Japan 2022
Awards: Winner, ASME IDETC International Essay Competition 2022 • GATE 2020 — 98.31 percentile

CERTIFICATIONS

CFA Level 1 CFA Institute